Binary MITM oracle ladder: sub-$2^{2\ell}$ holds at $\ell = 9$ and $\ell = 10$

2026-10-15. `research/binary_mitm_oracle_l9_l10_20261015/`; code `examples/binary_mitm_oracle_l8.rs` (rung arguments).
Class: engineering — the 2^{ℓ}-fold per-target saving moves with ℓ exactly as the design predicts; the $\Theta(2^n)$ relation-collection total and the ρ frontier are unchanged.

The row this extends

The binary `decomposition` ledger row landed its first sub-$2^{2\ell}$ oracle at $\ell = 8$ (PR 1919, merge `0f10f333`). That PR's next target was this ladder: the same group-law MITM pair-table design on fresh frozen instances at $\ell = 9$ and $\ell = 10$, holding zero disagreements and the counted sub-$2^{2\ell}$ property, with the table still cheap.

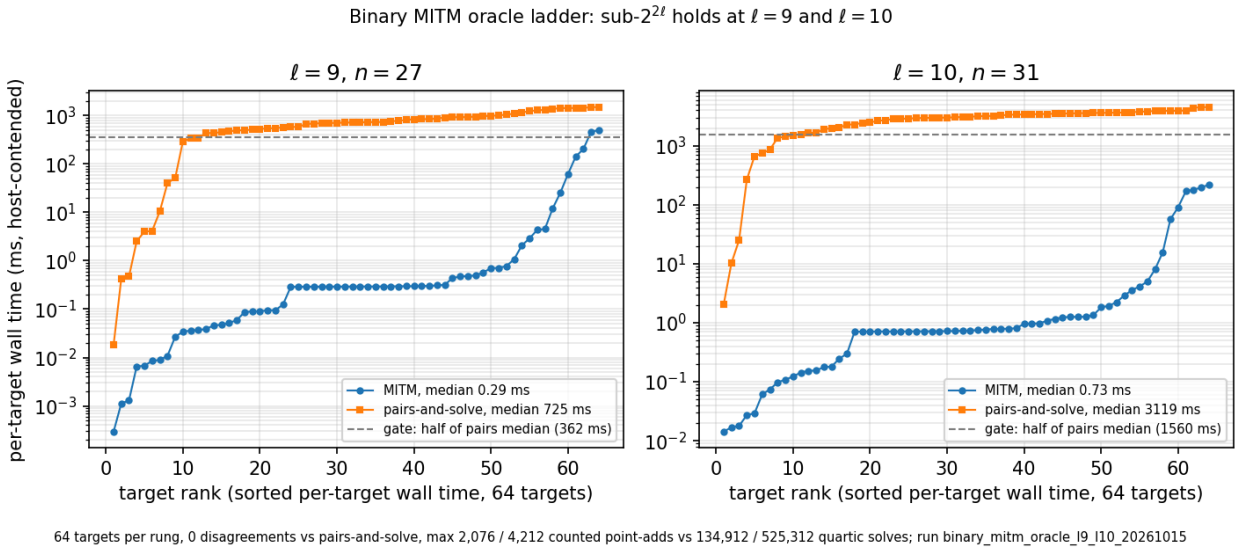
Instances (frozen per rung)

- Curve $y^2 + xy = x^3 + ax^2 + b$, $a = z \notin F_2$ (non-Koblitz), $b = 1$; deterministic target seed fixed in the example; 48 natural + 16 planted targets per rung.
- $\ell = 9$: $F_{2^{27}}$, base 519 points over $\langle 1, z, \ldots, z^8 \rangle$; table 134,594 sums, 4.1 MiB, 601.6 ms.
- $\ell = 10$: $F_{2^{31}}$, base 1,053 points over $\langle 1, z, \ldots, z^9 \rangle$; table 554,304 sums, 16.9 MiB, 4.8 s.
- Reference arm: `SubspaceOracle::low_order(ℓ, 1)` — Gaudry pairs-and-solve on the same subspace.

Results (shared arm64 macOS host, unpinned, contended by other sessions' builds)

ℓ	n	verdicts	disagr.	max MITM ops	median MITM	median pairs	ratio	gate
9	27	24/40	0	2,076	0.293 ms	724.9 ms	2,473.9x	PASS
10	31	20/44	0	4,212	0.735 ms	3,119.2 ms	4,244.2x	PASS

The counted-operation comparison is decisive: 2,076 / 4,212 point additions and lookups against 134,912 / 525,312 quartic solves — the 2^{ℓ}-fold gap holds at both rungs and the wall ratio widens because a quartic solve costs far more than a table lookup. Every witness is re-added in the group (asserted per target); MITM verdicts agree with pairs-and-solve on all 128 targets.



Per-target wall times of both arms at both rungs (sorted, log scale) with the half-of-pairs gate lines; run `binary_mitm_oracle_l9_l10_20261015`.

Controls

- The ℓ = 8 exhaustive control (complete triple enumeration at $n = 15$, $\ell = 5$, `cargo test --release --example binary_mitm_oracle_l8`) is unchanged and still passes.
- Per-run in-code assertions: every target on-curve, every witness group-verified, every verdict cross-checked against pairs-and-solve (0 disagreements, 128/128 targets).

Claim boundary

Stage `decomposition` records only. The $2^{2\ell}$-entry table replaces the $2^{2\ell}$ per-target scan, so relation collection stays $\Theta(2^n)$: no key recovery, no end-to-end ECDLP, no `vs_rho`, no asymptotic sub-ρ claim. FFD stays `unmeasured` with reason (group-law oracle). Wall times are L0 practicality notes under external build contention; the gate metric is the paired median ratio.

Next rung

$\ell = 11$ (table 2^{21} entries, 68 MiB) and $\ell = 12$ (the pairs-and-solve wall-clock ceiling of the reference arm), then the FFD-logged algebraic control on the same systems so the SAT/GB frontier comparison carries measured degree data.